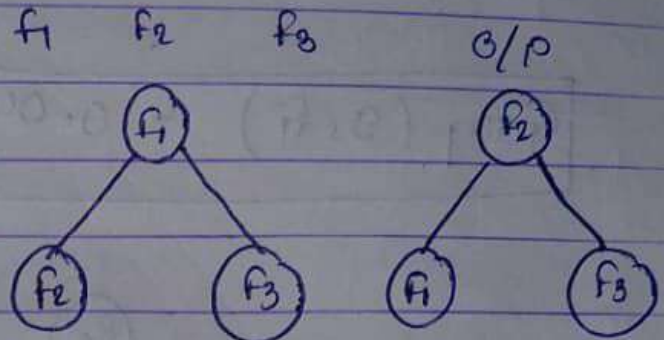
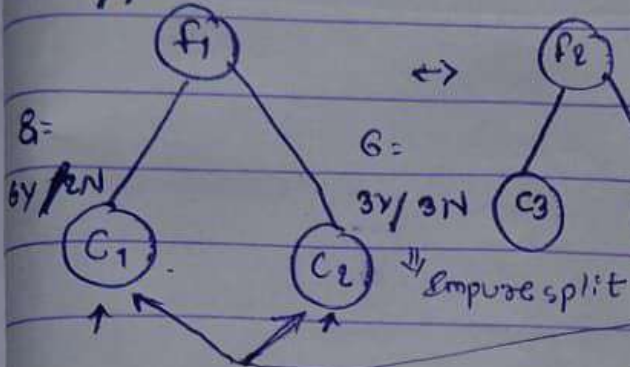


Information Gain

$$14 = 9y/5N$$



Entropy of the root node

$$\text{Gain}(S, f_1) = H(S) - \sum_{v \in \text{val}} \frac{|S_v|}{|S|} H(S_v)$$

Root Node

$$H(S) = -p + \log_2 p + -p - \log_2 p -$$

$$= -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$\approx 0.94$$

$$H(C_1) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8}$$

$$H(C_1) = 0.81$$

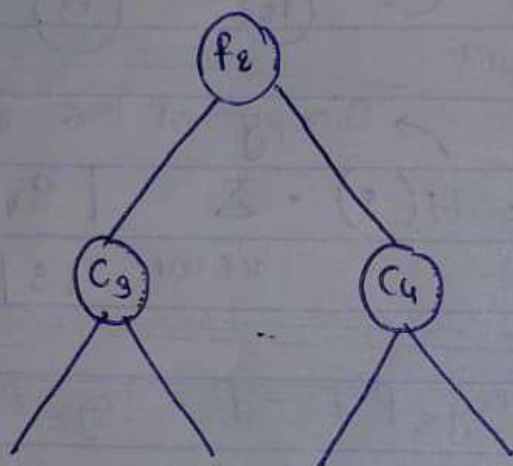
$$H(C_2) = 1$$

$\therefore S \rightarrow$ Entire sample from the category

$\therefore S_v \rightarrow$ After split many no. of yes and no

$$\text{Gain}(S, f_1) = 0.94 - \left[\frac{8}{14} \times 0.81 + \frac{6}{14} \times 1 \right]$$

$$\boxed{\text{Gain}(S, f_1) = 0.049}$$



$$\boxed{\text{Gain}(S, f_2) = 0.051}$$

$>$

$$\boxed{\text{Gain}(S, f_1) = 0.049}$$

Information ^{Gain} is Basically calculated.